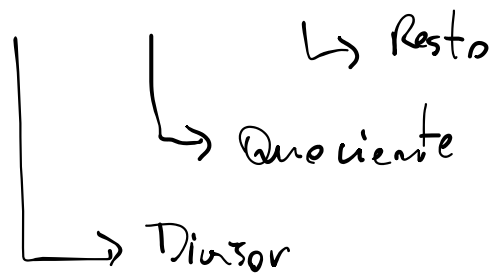


$$f(x) = x^3 + p \cdot x + q, \quad (x-a)^2$$

$$P(x) = D(x) \cdot Q(x) + R(x)$$



← Evandro

$$\partial P = n$$

$$P(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

$$= a_n (x - x_1)(x - x_2) \dots (x - x_n) \leftarrow$$

$$x_i, i = 1, 2, \dots, n$$

$$f(x) = (x-a)^2(x-b)$$

$$\begin{cases} f(a) = a^3 + p \cdot a + q = 0 & \dots (1) \end{cases}$$

$$\begin{cases} f'(a) = 3a^2 + p = 0 & \dots (2) \end{cases}$$

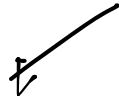
$$\begin{cases} p = -3a^2 & \dots (3) \end{cases}$$

$$\begin{cases} q = -a^3 - a \cdot p = -a^3 + 3a^3 = 2a^3 & \dots (4) \end{cases}$$

$$p^3 = -27a^6, \quad q^2 = 4a^6 \dots (5)$$

$$a^6 = \frac{q^2}{4} = -\frac{p^3}{27} \Rightarrow$$

$$\boxed{\frac{q^2}{4} + \frac{p^3}{27} = 0}$$



$$f'(x) = 2(x-a) \cdot 1 + (x-a)^2 \cdot 1$$

$$f'(x) = 2(x-a) + a^{-b} + (x-a) \cdot 1 \rightarrow 0$$

$$f'(x=a) = 2(\cancel{a-a})(a-b) + (\cancel{a-a})^2 = 0$$

$P(x)$, $\partial P = n$, a é raiz de multiplicidade k

$$P(x) = (x-a)^k Q(x), \quad Q(a) \neq 0$$

$$P^{(k-1)}(x=a) = 0$$

Novo problema:

$$f(x) = x^3 + px + q, \quad (x-a)(x-b)$$

$$f(x) \equiv (x-a)(x-b)(x-c)$$

$$\equiv [x^2 - (a+b)x + ab](x-c)$$

$$\equiv x^3 - (a+b+c)x^2 + (ab+ac+bc)x - abc$$

Newton - Girard

$$ax^2 + bx + c = 0, \quad x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2}$$

$$ax^2 + bx + c = a(x-x_1)(x-x_2)$$

$$x_1 + x_2 = -\frac{b}{a}$$

$$x_1 \cdot x_2 = \frac{c}{a}$$

$$x^3 + px + q \equiv x^3 - (a+b+c)x^2 + (ab+ac+bc)x - abc$$

$$x^3 + px + q = x^3 - (a+b+c)x^2 + (ab+ac+bc)x - abc$$

$$\begin{cases} a+b+c=0 \\ ab+bc+ac=p \\ -abc=q \end{cases}$$

$$k = \frac{b}{a} \Rightarrow \boxed{b = k \cdot a}$$

$$\begin{cases} c = -(1+k)a \\ p = ka^2 + ka \cdot [-(1+k)a] - a^2(1+k) \leftarrow \\ q = -a(ka) [-(1+k)a] = \boxed{k(1+k)a^3 = q} \end{cases}$$

$$p = ka^2 - k(1+k)a^2 - a^2(1+k)$$

$$p = \cancel{ka^2} - \cancel{ka^2} - k^2a^2 - a^2 - ka^2$$

$$\boxed{p = -(1+k+k^2)a^2}$$

$$\begin{cases} p^3 = -(1+k+k^2)^3 a^6 \\ q^2 = k^2(1+k)^2 a^6 \end{cases}$$

$$a^6 = \frac{q^2}{k^2(1+k)^2} = -\frac{p^3}{(1+k+k^2)^3}$$

$$\boxed{\frac{q^2}{k^2(1+k)^2} + \frac{p^3}{(1+k+k^2)^3} = 0}$$

$$a=b \Rightarrow k=1 \Rightarrow \frac{q^2}{4} + \frac{p^3}{27} = 0$$